Atiyah–Macdonald, Chapter 9: Discrete Valuation Rings and Dedekind Domains

Solutions to the exercises

Conventions. K denotes the field of fractions of the domain A. A *Dedekind domain* is a Noetherian integrally closed domain of dimension one (Thm. 9.3); in particular it is not a field, every non-zero prime is maximal, and $A_{\mathfrak{p}}$ is a discrete valuation ring (DVR) for every non-zero prime $\mathfrak{p}$. From the text I use: 9.2 (a Noetherian local domain of dimension one with principal maximal ideal is a DVR; in a DVR with maximal ideal (π) the non-zero ideals are exactly (π^k), $k \geq 0$); 9.3, 9.4 (unique factorization of non-zero ideals into primes); 9.6–9.9 (a non-zero fractional ideal of a Dedekind domain is invertible; the non-zero fractional ideals form a group I, free on the non-zero primes; $H = I/P$ is the ideal class group); the localization facts of Ch. 3 (3.3, 3.4, 3.8, 3.9, 3.10, 3.11); integral closure commutes with localization (5.12); the Chinese remainder theorem (1.10, 1.16); and the structure theorem for finitely generated modules over a principal ideal domain, including uniqueness of the elementary divisors.

A recurring principle (3.8/3.9): *if $X \subseteq Y$ are ideals (or modules) with $X_{\mathfrak{m}} = Y_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$, then $X = Y$* (apply 3.8 to Y/X, using exactness of localization). In a Dedekind domain the local rings $A_{\mathfrak{m}}$ at maximal ideals are DVRs, so identities can be verified in a DVR.

This chapter has no numbered in-text exercises (the text only says "check that v is well defined" in the proof of 9.2), and the 9 end-of-chapter exercises are all solved below.

End-of-chapter exercises

Exercise 1

> **Problem.** Let A be a Dedekind domain, S a multiplicatively closed subset of A. Show that $S^{-1}A$ is either a Dedekind domain or the field of fractions of A.
>
> Suppose that $S \neq A - \{0\}$, and let H, H' be the ideal class groups of A and $S^{-1}A$ respectively. Show that extension of ideals induces a surjective homomorphism $H \to H'$.

Solution. We assume $0 \notin S$ (otherwise $S^{-1}A = 0$ is not a domain). Then $A \subseteq S^{-1}A \subseteq K$ and $S^{-1}A$ is a domain with field of fractions K.

Noetherian. Every ideal $\mathfrak{b}$ of $S^{-1}A$ is extended, $\mathfrak{b} = (\mathfrak{b}^c)^e$ (3.11(i)); if $\mathfrak{b}^c$ is generated by $x_1, \ldots, x_n$, then $\mathfrak{b}$ is generated by $x_1/1, \ldots, x_n/1$.

Integrally closed. A is integrally closed, so $S^{-1}A$ is integrally closed in K (5.12).

Primes. The primes of $S^{-1}A$ correspond bijectively, preserving inclusion, to the primes of A not meeting S (3.11(iv)). The prime (0) always corresponds to (0) ($0 \cap S = \emptyset$), and every non-zero prime of A is maximal. Hence every chain of primes of $S^{-1}A$ has length at most 1.

Two cases. If every non-zero prime of A meets S, then (0) is the only prime of $S^{-1}A$; a non-unit of $S^{-1}A$ lies in some maximal ideal, which is a non-zero prime unless the element is 0; so every non-zero element is a unit and $S^{-1}A = K$. Otherwise $S^{-1}A$ has a non-zero prime, so $\dim S^{-1}A = 1$, and $S^{-1}A$ is a Noetherian, integrally closed domain of dimension one, i.e. a Dedekind domain.

The map $H \to H'$. (If $S^{-1}A = K$ the group H' is trivial, since the only non-zero fractional ideal of K is K, and there is nothing to prove; so assume $S^{-1}A$ is Dedekind.) Let I, I' be the groups of non-zero fractional ideals of A and $S^{-1}A$. For $M \in I$ put $\theta(M) = S^{-1}M = S^{-1}A \cdot M \subseteq K$. If $xM \subseteq A$ with $0 \neq x \in A$ then $x\,S^{-1}M \subseteq S^{-1}A$, so $\theta(M)$ is a non-zero fractional ideal. Clearly $\theta(MN) = \theta(M)\theta(N)$ (products are generated by the products of generators), so $\theta : I \to I'$ is a group homomorphism (it preserves identities and inverses because $\theta(A) = S^{-1}A$). For $u \in K^*$, $\theta((u)) = (u)$, so θ maps the principal fractional ideals P into P' and induces $H = I/P \to H' = I'/P'$.

Surjective. It suffices to show $\theta : I \to I'$ is surjective. Let $N' \in I'$, and choose $0 \neq z \in S^{-1}A$ with $zN' \subseteq S^{-1}A$; writing $z = a/s$ we may replace z by $a = sz \in A - \{0\}$ (s is a unit of $S^{-1}A$). Then $\mathfrak{b} = aN'$ is a non-zero ideal of $S^{-1}A$, hence $\mathfrak{b} = \mathfrak{a}^e = S^{-1}\mathfrak{a}$ with $\mathfrak{a} = \mathfrak{b}^c$ an ideal of A (3.11(i)). Therefore $N' = a^{-1}S^{-1}\mathfrak{a} = S^{-1}(a^{-1}\mathfrak{a})$, and $M = a^{-1}\mathfrak{a}$ is a non-zero fractional ideal of A ($aM = \mathfrak{a} \subseteq A$). So $N' = \theta(M)$. □

Exercise 2

> **Problem.** Let A be a Dedekind domain. If $f = a_0 + a_1x + \cdots + a_nx^n$ is a polynomial with coefficients in A, the *content* of f is the ideal $c(f) = (a_0, \ldots, a_n)$ in A. Prove *Gauss's lemma* that $c(fg) = c(f)c(g)$. [Localize at each maximal ideal.]

Solution. Always $c(fg) \subseteq c(f)c(g)$, because every coefficient of fg is a sum of products a_ib_j of coefficients of f and g. By the principle in the Conventions it suffices to show $c(fg)_{\mathfrak{m}} = (c(f)c(g))_{\mathfrak{m}}$ for every maximal ideal $\mathfrak{m}$. Since localization commutes with products (3.4), $(c(f)c(g))_{\mathfrak{m}} = c(f)_{\mathfrak{m}}c(g)_{\mathfrak{m}}$, and $c(f)_{\mathfrak{m}}$ is the ideal of $A_{\mathfrak{m}}$ generated by the (images of the) coefficients of f, i.e. the content of f viewed in $A_{\mathfrak{m}}[x]$; likewise for g and fg. So we may replace A by $A_{\mathfrak{m}}$ and assume A is a DVR with maximal ideal (π) and valuation v (recall $\mathfrak{m} \neq 0$ since A is not a field).

If $f = 0$ or $g = 0$ both sides are 0. Otherwise let $a = \min_i v(a_i)$ and $b = \min_j v(b_j)$. Then $c(f) = (\pi^a)$, $c(g) = (\pi^b)$ (ideals of a DVR are the (π^k) and π^a divides every coefficient). Write $f = \pi^a f_1$, $g = \pi^b g_1$ with $f_1, g_1 \in A[x]$; then some coefficient of f_1 and some coefficient of g_1 is a unit, so $\bar{f}_1 \neq 0$ and $\bar{g}_1 \neq 0$ in $(A/\mathfrak{m})[x]$. As $(A/\mathfrak{m})[x]$ is a domain, $\bar{f}_1\bar{g}_1 \neq 0$, i.e. some coefficient of f_1g_1 is not in $\mathfrak{m}$, so is a unit, and $c(f_1g_1) = (1)$. Hence $c(fg) = \pi^{a+b}c(f_1g_1) = (\pi^{a+b}) = c(f)c(g)$. □

Exercise 3

Problem. A valuation ring (other than a field) is Noetherian if and only if it is a discrete valuation ring.

Solution. Recall that a *valuation ring* is an integral domain A with field of fractions K such that for every $x \in K^*$ either $x \in A$ or $x^{-1} \in A$.

($\Leftarrow$) A DVR is Noetherian: its ideals are 0 and $\mathfrak{m}^k = (x^k)$, all finitely generated (this is shown in the text before 9.2).

($\Rightarrow$) Let A be a Noetherian valuation ring which is not a field.

A is local. Let x, y be non-units of A, not both 0, say $y \neq 0$. Then $x/y \in A$ or $y/x \in A$; by symmetry (if $x \neq 0$) suppose $a = x/y \in A$. Then $x + y = y(1 + a)$ is a multiple of the non-unit y, hence a non-unit. Also ax is a non-unit for $a \in A$, x a non-unit. So the non-units form an ideal, which is therefore the unique maximal ideal $\mathfrak{m}$, and $\mathfrak{m} \neq 0$ because A is not a field.

Finitely generated ideals are principal. For $a, b \in A$ not both 0 say $b \neq 0$ and $a/b \in A$ (else $b/a \in A$ and swap the roles); then $a \in (b)$ and $(a, b) = (b)$. By induction every finitely generated ideal of A is principal. A is Noetherian, so A is a principal ideal domain.

Dimension one. A non-zero prime ideal in a PID is maximal, so the only non-zero prime is $\mathfrak{m}$, and $\dim A = 1$.

Thus A is a Noetherian local domain of dimension one whose maximal ideal is principal, so A is a DVR by 9.2 (iii)$\Rightarrow$(i). $\square$

Exercise 4

Problem. Let A be a local domain which is not a field and in which the maximal ideal $\mathfrak{m}$ is principal and $\bigcap_{n=1}^{\infty} \mathfrak{m}^n = 0$. Prove that A is a discrete valuation ring.

Solution. Let $\mathfrak{m} = (x)$; $x \neq 0$ since A is not a field. Let $a \in A$, $a \neq 0$. As $\bigcap \mathfrak{m}^n = 0$ there is a largest $k \geq 0$ with $a \in \mathfrak{m}^k = (x^k)$; write $a = x^k u$. Then $u \notin \mathfrak{m}$ (else $a \in (x^{k+1})$), so u is a unit. The integer $k =: v(a)$ is uniquely determined by a: if $x^k u = x^l w$ with $k < l$ and u, w units, cancelling x^k in the domain gives $u = x^{l-k} w \in \mathfrak{m}$, absurd.

Every element of K^* is a/b with $a, b \in A - \{0\}$; define $v(a/b) = v(a) - v(b)$. This is well defined: if $a/b = a'/b'$ then $ab' = a'b$, so $v(a) + v(b') = v(a') + v(b)$ (units do not change v, and $v(ab) = v(a) + v(b)$ on $A - \{0\}$ since $x^k u \cdot x^l w = x^{k+l} uw$). Thus $v : K^* \to \mathbf{Z}$ is a homomorphism, and it is onto because $v(x) = 1$.

Ultrametric inequality. Let $\alpha = a/c$, $\beta = b/c$ with common denominator c (put $v(0) = +\infty$; if $\alpha + \beta = 0$, or if α or β is 0, there is nothing to prove). Say $v(a) \leq v(b)$ and $k = v(a)$. Then $a, b \in (x^k)$, so $a + b \in (x^k)$ and $v(a + b) \geq k$ (if $a + b \neq 0$). Hence $v(\alpha + \beta) = v(a + b) - v(c) \geq v(a) - v(c) = \min(v(\alpha), v(\beta))$. So v is a discrete valuation.

A is its valuation ring. If $z = a/b$ with $v(z) \geq 0$, i.e. $v(a) \geq v(b)$, then $a = x^k u$, $b = x^l w$ with $k \geq l$, so $z = x^{k-l} uw^{-1} \in A$. Conversely for $a \in A - \{0\}$, $v(a) \geq 0$. Hence $A = \{z : v(z) \geq 0\} \cup (0)$ is the valuation ring of v, and A is a discrete valuation ring by definition. $\square$

Exercise 5

Problem. Let M be a finitely generated module over a Dedekind domain. Prove that M is flat $\iff$ M is torsion-free. [Use Chapter 3, Exercise 13 and Chapter 7, Exercise 16.]

Solution. Let A be the Dedekind domain. (I give a self-contained argument that does not use Chapter 7, Exercise 16; I only use the structure theorem for finitely generated modules over a PID.)

($\Rightarrow$) Let M be flat and $0 \neq a \in A$. Multiplication by a, $A \to A$, is injective; tensoring with the flat module M gives that $M \to M$, $m \mapsto am$, is injective. So no non-zero element of M is killed by a non-zero element of A: M is torsion-free. (This direction holds for any domain and any module.)

($\Leftarrow$) Let M be finitely generated and torsion-free. By Ch. 3, Ex. 13, $M_{\mathfrak{m}}$ is torsion-free for every maximal ideal $\mathfrak{m}$. Here $\mathfrak{m} \neq 0$ and $A_{\mathfrak{m}}$ is a DVR, hence a PID. $M_{\mathfrak{m}}$ is a finitely generated torsion-free module over the PID $A_{\mathfrak{m}}$, hence free (structure theorem: $M_{\mathfrak{m}} \cong A_{\mathfrak{m}}^r \oplus T$ with T the torsion submodule, and $T = 0$). A free module is flat. Thus $M_{\mathfrak{m}}$ is a flat $A_{\mathfrak{m}}$-module for all maximal ideals $\mathfrak{m}$, and so M is flat over A by 3.10. $\square$

Exercise 6

Problem. Let M be a finitely generated torsion module ($T(M) = M$) over a Dedekind domain A. Prove that M is uniquely representable as a finite direct sum of modules $A/\mathfrak{p}_i^{n_i}$, where $\mathfrak{p}_i$ are non-zero prime ideals of A. [For each $\mathfrak{p} \neq 0$, $M_{\mathfrak{p}}$ is a torsion $A_{\mathfrak{p}}$-module; use the structure theorem for modules over a principal ideal domain.]

Solution. "Uniquely" means: the multiset of pairs $(\mathfrak{p}_i, n_i)$ (with $n_i \geq 1$) is determined by M.

$\mathrm{Ann}(M) \neq 0$. If $x_1, \ldots, x_n$ generate M and $0 \neq a_i \in \mathrm{Ann}(x_i)$, then $a = a_1 \cdots a_n \neq 0$ kills M. Let $\mathfrak{a} = \mathrm{Ann}(M) \neq 0$. If $\mathfrak{a} = A$ then $M = 0$ (the empty sum); otherwise by 9.4 $\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$ with distinct non-zero primes $\mathfrak{p}_i$ and $e_i \geq 1$. Put $\mathfrak{q}_i = \mathfrak{p}_i^{e_i}$ and $\mathfrak{b}_i = \prod_{j \neq i} \mathfrak{q}_j$.

Existence. The ideals $\mathfrak{q}_i$ are pairwise coprime (distinct maximal ideals have coprime powers: a maximal ideal containing $\mathfrak{q}_i + \mathfrak{q}_j$ would contain $\mathfrak{p}_i$ and $\mathfrak{p}_j$, forcing $\mathfrak{p}_i = \mathfrak{p}_j$), so $\mathfrak{b}_1 + \cdots + \mathfrak{b}_r = (1)$ (a maximal ideal containing all $\mathfrak{b}_i$ would, for $r \geq 2$, contain some $\mathfrak{q}_j$, hence equal some $\mathfrak{p}_j$ with $j \neq 1$, and also some $\mathfrak{q}_k$, $k \neq j$, hence equal $\mathfrak{p}_k \neq \mathfrak{p}_j$; for $r = 1$, $\mathfrak{b}_1 = A$). Write $1 = \sum c_i b_i$ with $b_i \in \mathfrak{b}_i$, and put $M_i = \{x \in M : \mathfrak{q}_i x = 0\}$. For $x \in M$, $x = \sum c_i b_i x$ and $c_i b_i x \in M_i$ because $\mathfrak{q}_i b_i \subseteq \mathfrak{a}$. The sum is direct: if $\sum y_i = 0$ with $y_i \in M_i$, multiply by b_i: $b_i y_j = 0$ for $j \neq i$ (since $\mathfrak{q}_j \mid \mathfrak{b}_i$ there) so $b_i y_i = 0$; as also $\mathfrak{q}_i y_i = 0$ and $\mathfrak{b}_i + \mathfrak{q}_i = A$ ($\mathfrak{q}_i$ is coprime to $\mathfrak{b}_i$), $y_i = 0$. So $M = \bigoplus_{i=1}^r M_i$, each M_i finitely generated (a quotient of M) and killed by $\mathfrak{q}_i = \mathfrak{p}_i^{e_i}$.

Fix i and write $\mathfrak{p} = \mathfrak{p}_i$, $\mathfrak{q} = \mathfrak{q}_i$. For $s \in A - \mathfrak{p}$ we have $(s) + \mathfrak{q} = A$ (the only maximal ideal containing $\mathfrak{q}$ is $\mathfrak{p}$), so $ts \equiv 1 \pmod{\mathfrak{q}}$ for some t, and s acts bijectively on M_i. Hence $M_i \to (M_i)_{\mathfrak{p}}$, $x \mapsto x/1$, is an isomorphism of A-modules (injective since $sx = 0 \Rightarrow x = tsx = 0$; surjective since $x/s = tx/1$). Now $(M_i)_{\mathfrak{p}}$ is a finitely generated module over the DVR $A_{\mathfrak{p}}$, killed by $\mathfrak{p}^e A_{\mathfrak{p}}$, so it is a torsion module and by the structure theorem

$$(M_i)_{\mathfrak{p}} \cong \bigoplus_j A_{\mathfrak{p}}/\mathfrak{p}^{k_j} A_{\mathfrak{p}} \qquad (k_j \geq 1),$$

and $A_{\mathfrak{p}}/\mathfrak{p}^k A_{\mathfrak{p}} \cong (A/\mathfrak{p}^k)_{\mathfrak{p}} \cong A/\mathfrak{p}^k$ (by the same argument as above, elements outside $\mathfrak{p}$ act invertibly on $A/\mathfrak{p}^k$). Therefore $M_i \cong \bigoplus_j A/\mathfrak{p}_i^{k_{ij}}$ as A-modules, and $M \cong \bigoplus_{i,j} A/\mathfrak{p}_i^{k_{ij}}$.

Uniqueness. Suppose $M \cong \bigoplus_{j=1}^{m} A/\mathfrak{p}_j^{n_j}$ with non-zero primes $\mathfrak{p}_j$ and $n_j \geq 1$. Fix a non-zero prime $\mathfrak{p}$. If $\mathfrak{p}_j \neq \mathfrak{p}$, then $\mathfrak{p}_j^{n_j} \not\subseteq \mathfrak{p}$ (else $\mathfrak{p}_j \subseteq \mathfrak{p}$, so $\mathfrak{p}_j = \mathfrak{p}$ as both are maximal), so there is $s \in \mathfrak{p}_j^{n_j} - \mathfrak{p}$, which kills $A/\mathfrak{p}_j^{n_j}$, and hence $(A/\mathfrak{p}_j^{n_j})_{\mathfrak{p}} = 0$. If $\mathfrak{p}_j = \mathfrak{p}$ then $(A/\mathfrak{p}^{n_j})_{\mathfrak{p}} \cong A_{\mathfrak{p}}/\mathfrak{p}^{n_j}A_{\mathfrak{p}}$ (3.4). Thus

$$M_{\mathfrak{p}} \cong \bigoplus_{j:\ \mathfrak{p}_j=\mathfrak{p}} A_{\mathfrak{p}}/\mathfrak{p}^{n_j}A_{\mathfrak{p}}.$$

By the uniqueness part of the structure theorem over the PID $A_{\mathfrak{p}}$ (the elementary divisors $\mathfrak{p}^{n_j}A_{\mathfrak{p}}$, equivalently the exponents n_j, are determined by $M_{\mathfrak{p}}$), the multiset $\{n_j : \mathfrak{p}_j = \mathfrak{p}\}$ depends only on M. This holds for every non-zero prime $\mathfrak{p}$, so the multiset of pairs $(\mathfrak{p}_j, n_j)$ is determined by M. □

Exercise 7

> **Problem.** Let A be a Dedekind domain and $\mathfrak{a} \neq 0$ an ideal in A. Show that every ideal in $A/\mathfrak{a}$ is principal.
>
> Deduce that every ideal in A can be generated by at most 2 elements.

Solution. Let $\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$ (9.4; if $\mathfrak{a} = A$ then $A/\mathfrak{a} = 0$ and there is nothing to prove). Ideals of $A/\mathfrak{a}$ are of the form $\mathfrak{b}/\mathfrak{a}$ with $\mathfrak{b} \supseteq \mathfrak{a}$ an ideal of A. If $\mathfrak{b} \supseteq \mathfrak{a}$ then $\mathfrak{b} \neq 0$ and $\mathfrak{b}^{-1}\mathfrak{a} \subseteq \mathfrak{b}^{-1}\mathfrak{b} = A$, so $\mathfrak{a} = \mathfrak{b} \cdot (\mathfrak{b}^{-1}\mathfrak{a})$ is a multiple of $\mathfrak{b}$; by unique factorization $\mathfrak{b} = \mathfrak{p}_1^{f_1} \cdots \mathfrak{p}_r^{f_r}$ with $0 \leq f_i \leq e_i$ (only primes dividing $\mathfrak{a}$ occur in $\mathfrak{b}$).

Choice of a generator. For each i, $\mathfrak{p}_i^{f_i} \neq \mathfrak{p}_i^{f_i+1}$ (unique factorization), so pick $x_i \in \mathfrak{p}_i^{f_i} - \mathfrak{p}_i^{f_i+1}$. The ideals $\mathfrak{p}_i^{f_i+1}$ are pairwise coprime, so by the Chinese remainder theorem (1.10) there is $x \in A$ with $x \equiv x_i \pmod{\mathfrak{p}_i^{f_i+1}}$ for all i. Then $x \in \mathfrak{p}_i^{f_i}$ and $x \notin \mathfrak{p}_i^{f_i+1}$ (since $x - x_i \in \mathfrak{p}_i^{f_i+1}$ and $x_i \in \mathfrak{p}_i^{f_i} - \mathfrak{p}_i^{f_i+1}$).

Claim: $\mathfrak{b} = (x) + \mathfrak{a}$. Both are ideals, so we compare localizations at every maximal ideal $\mathfrak{m}$ (Conventions). Recall that $\mathfrak{p}^{k+1}A_{\mathfrak{p}} \cap A = \mathfrak{p}^{k+1}$ for a non-zero prime $\mathfrak{p}$ (the ideal $\mathfrak{p}^{k+1}$ is $\mathfrak{p}$-primary, 4.2, and 4.8(ii)).

- $\mathfrak{m} = \mathfrak{p}_i$: in the DVR $A_{\mathfrak{p}_i}$ the element x lies in $\mathfrak{p}_i^{f_i}A_{\mathfrak{p}_i}$ but not in $\mathfrak{p}_i^{f_i+1}A_{\mathfrak{p}_i}$ (otherwise $x \in \mathfrak{p}_i^{f_i+1}A_{\mathfrak{p}_i} \cap A = \mathfrak{p}_i^{f_i+1}$), so $(x)_{\mathfrak{m}} = \mathfrak{p}_i^{f_i}A_{\mathfrak{m}}$. Moreover $\mathfrak{a}_{\mathfrak{m}} = \mathfrak{p}_i^{e_i}A_{\mathfrak{m}}$ ($\mathfrak{p}_j \not\subseteq \mathfrak{m}$ for $j \neq i$, so $\mathfrak{p}_j$ becomes the unit ideal), and $\mathfrak{p}_i^{e_i}A_{\mathfrak{m}} \subseteq \mathfrak{p}_i^{f_i}A_{\mathfrak{m}}$. So $((x) + \mathfrak{a})_{\mathfrak{m}} = \mathfrak{p}_i^{f_i}A_{\mathfrak{m}} = \mathfrak{b}_{\mathfrak{m}}$.
- $\mathfrak{m} \notin \{\mathfrak{p}_1, \ldots, \mathfrak{p}_r\}$: then $\mathfrak{a}_{\mathfrak{m}} = A_{\mathfrak{m}}$ and $\mathfrak{b}_{\mathfrak{m}} = A_{\mathfrak{m}}$ (as $\mathfrak{b} \supseteq \mathfrak{a}$), so both sides equal $A_{\mathfrak{m}}$.

Thus $\mathfrak{b} = (x) + \mathfrak{a}$, i.e. $\mathfrak{b}/\mathfrak{a}$ is generated by the class of x in $A/\mathfrak{a}$. Every ideal of $A/\mathfrak{a}$ is principal.

Every ideal is 2-generated. Let $\mathfrak{b}$ be an ideal of A. If $\mathfrak{b} = 0$ it is principal. Otherwise pick $0 \neq a \in \mathfrak{b}$ and apply the above to $\mathfrak{a} = (a) \neq 0$: the ideal $\mathfrak{b}/(a)$ of $A/(a)$ is principal, generated by the class of some $x \in \mathfrak{b}$; hence $\mathfrak{b} = (a, x)$. □

Exercise 8

Problem. Let $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}$ be three ideals in a Dedekind domain. Prove that

$$\mathfrak{a} \cap (\mathfrak{b} + \mathfrak{c}) = (\mathfrak{a} \cap \mathfrak{b}) + (\mathfrak{a} \cap \mathfrak{c}),$$
$$\mathfrak{a} + (\mathfrak{b} \cap \mathfrak{c}) = (\mathfrak{a} + \mathfrak{b}) \cap (\mathfrak{a} + \mathfrak{c}).$$

[Localize.]

Solution. In each identity the right-hand side is contained in the left-hand side for the first, and the left-hand side is contained in the right-hand side for the second, in any ring: $(\mathfrak{a}\cap\mathfrak{b})+(\mathfrak{a}\cap\mathfrak{c}) \subseteq \mathfrak{a} \cap (\mathfrak{b} + \mathfrak{c})$, and $\mathfrak{a} + (\mathfrak{b} \cap \mathfrak{c}) \subseteq (\mathfrak{a} + \mathfrak{b}) \cap (\mathfrak{a} + \mathfrak{c})$. So it suffices to compare localizations at every maximal ideal $\mathfrak{m}$ (Conventions). Localization commutes with finite sums and intersections (3.4), so we must check both identities for the three ideals $\mathfrak{a}_\mathfrak{m}, \mathfrak{b}_\mathfrak{m}, \mathfrak{c}_\mathfrak{m}$ of the DVR $A_\mathfrak{m}$.

In a DVR with maximal ideal (π) the ideals are (π^k) $(k \geq 0)$ and 0, so any two ideals are comparable under inclusion. By symmetry in $\mathfrak{b}, \mathfrak{c}$ we may assume $\mathfrak{b} \subseteq \mathfrak{c}$. Then $\mathfrak{b} + \mathfrak{c} = \mathfrak{c}$ and $\mathfrak{b} \cap \mathfrak{c} = \mathfrak{b}$. First identity: $\mathfrak{a} \cap (\mathfrak{b} + \mathfrak{c}) = \mathfrak{a} \cap \mathfrak{c}$, and $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{c}$ gives $(\mathfrak{a} \cap \mathfrak{b}) + (\mathfrak{a} \cap \mathfrak{c}) = \mathfrak{a} \cap \mathfrak{c}$. Second identity: $\mathfrak{a} + (\mathfrak{b} \cap \mathfrak{c}) = \mathfrak{a} + \mathfrak{b}$, and $\mathfrak{a} + \mathfrak{b} \subseteq \mathfrak{a} + \mathfrak{c}$ gives $(\mathfrak{a} + \mathfrak{b}) \cap (\mathfrak{a} + \mathfrak{c}) = \mathfrak{a} + \mathfrak{b}$. So both identities hold locally, hence globally. □

Exercise 9

Problem. (*Chinese Remainder Theorem.*) Let $\mathfrak{a}_1, \ldots, \mathfrak{a}_n$ be ideals and let $x_1, \ldots, x_n$ be elements in a Dedekind domain A. Then the system of congruences $x \equiv x_i \pmod{\mathfrak{a}_i}$ $(1 \leq i \leq n)$ has a solution x in $A \iff x_i \equiv x_j \pmod{\mathfrak{a}_i + \mathfrak{a}_j}$ whenever $i \neq j$.

[This is equivalent to saying that the sequence of A-modules

$$A \xrightarrow{\phi} \bigoplus_{i=1}^{n} A/\mathfrak{a}_i \xrightarrow{\psi} \bigoplus_{i<j} A/(\mathfrak{a}_i + \mathfrak{a}_j)$$

is exact, where ϕ and ψ are defined as follows: $\phi(x) = (x+\mathfrak{a}_1, \ldots, x+\mathfrak{a}_n)$; $\psi(x_1+\mathfrak{a}_1, \ldots, x_n+\mathfrak{a}_n)$ has (i,j)-component $x_i - x_j + \mathfrak{a}_i + \mathfrak{a}_j$. To show that this sequence is exact it is enough to show that it is exact when localized at any $\mathfrak{p} \neq 0$: in other words we may assume that A is a discrete valuation ring, and then it is easy.]

Solution. *The reformulation.* A tuple $(x_i + \mathfrak{a}_i)_i$ lies in $\operatorname{Ker}\psi$ iff $x_i - x_j \in \mathfrak{a}_i + \mathfrak{a}_j$ for all $i < j$, and lies in $\operatorname{Im}\phi$ iff there is $x \in A$ with $x \equiv x_i \pmod{\mathfrak{a}_i}$ for all i. So the statement says exactly that $\operatorname{Ker}\psi = \operatorname{Im}\phi$. Always $\psi\phi = 0$: the (i,j)-component of $\psi\phi(x)$ is $x - x$ modulo $\mathfrak{a}_i + \mathfrak{a}_j$. (This is also the easy direction of the equivalence: a solution x gives $x_i - x_j \equiv x - x = 0$.) Put $E = \operatorname{Ker}\psi / \operatorname{Im}\phi$; we must show $E = 0$.

Localization. Localization is exact (3.3), so for every maximal ideal $\mathfrak{m}$ the localized sequence is exact in the middle iff $E_\mathfrak{m} = 0$ (because $(\operatorname{Ker}\psi)_\mathfrak{m} = \operatorname{Ker}\psi_\mathfrak{m}$ and $(\operatorname{Im}\phi)_\mathfrak{m} = \operatorname{Im}\phi_\mathfrak{m}$). Using $(A/\mathfrak{a})_\mathfrak{m} \cong A_\mathfrak{m}/\mathfrak{a}_\mathfrak{m}$ and $(\mathfrak{a}_i+\mathfrak{a}_j)_\mathfrak{m} = (\mathfrak{a}_i)_\mathfrak{m}+(\mathfrak{a}_j)_\mathfrak{m}$ (3.4), the localized sequence is the same sequence for the ring $A_\mathfrak{m}$ and the ideals $(\mathfrak{a}_i)_\mathfrak{m}$. If $E_\mathfrak{m} = 0$ for all maximal $\mathfrak{m}$, then $E = 0$ (3.8). Every maximal ideal $\mathfrak{m}$ is non-zero, so $A_\mathfrak{m}$ is a DVR.

The DVR case. Let A be a DVR with maximal ideal (π). Each $\mathfrak{a}_i$ is 0 or (π^{k_i}); write $\mathfrak{a}_i = (\pi^{k_i})$ with $k_i \in \{0, 1, 2, \ldots\} \cup \{\infty\}$ $(\pi^\infty := 0)$. Then $\mathfrak{a}_i + \mathfrak{a}_j = (\pi^{\min(k_i,k_j)})$. Suppose $x_i - x_j \in \mathfrak{a}_i + \mathfrak{a}_j$

for all i, j. Renumber so that $k_1 \le k_2 \le \cdots \le k_n$ and take $x = x_n$. For $i \le n$, $x_n - x_i \in \mathfrak{a}_i + \mathfrak{a}_n = (\pi^{\min(k_i,k_n)}) = (\pi^{k_i}) = \mathfrak{a}_i$ (for $i = n$ this is 0). So $x \equiv x_i \pmod{\mathfrak{a}_i}$ for all i: a solution exists.

Hence $E_{\mathfrak{m}} = 0$ for every maximal $\mathfrak{m}$, so $E = 0$ and the system is solvable iff the compatibility conditions hold. $\square$